



LESSON 1.3

MULTIPLYING FRACTIONS

LESSON INTRODUCTION

MA.5.AR.1.2 Solve real-world problems involving the addition, subtraction, or multiplication of fractions, including mixed numbers and fractions greater than 1.

LEARNING TARGETS

- I can solve real-world problems involving multiplication of fractions.

BENCHMARK COHERENCE

WORKING TOWARD

MA.6.NSO.2.3 Solve multistep real-world problems involving any of the four operations with positive multidigit decimals or positive fractions, including mixed numbers.

ESSENTIAL QUESTIONS

- What is an algorithm?
- How can you use an area model or array model to represent and solve for the product of fractions?
- How does the algorithm for multiplying fractions connect to the area model?
- What is the process for multiplying mixed numbers?

LESSON NARRATIVE

In this lesson, students are guided to create and use visual models to show how to multiply a fraction by a whole number and a fraction by a fraction. Students then utilize their understanding of the visual model to support their understanding of the area model for multiplying mixed numbers with the algorithm.

COMPONENT SUMMARY

1.3.1 WARM-UP (~5 MIN.)

Students identify and extend a pattern using expressions

1.3.3 GUIDED PRACTICE (10-15 MIN.)

Multiplying fractions practice (SE p. 17)

1.3.5 WRAP-UP (~5 MIN.)

Assessment on multiplying fractions and mixed numbers

1.3.2 GUIDED INSTRUCTION (15-25 MIN.)

Use array and area models to multiply fractions in the context of a recipe

1.3.4 COLLABORATION (10 MIN.)

Students generalize fraction multiplication from arrays to create mixed number to improper fraction algorithms

RESOURCES

GLOSSARY

Key Terms:

- Algorithm

Additional Terms:

- Array, Area model
- Numerator, Denominator
- Mixed number

- (Optional) Graph paper
- (Optional) Math manipulatives (counters, fraction circles, fraction bars, etc.)

- (Optional) Gather math manipulatives to represent fraction multiplication and visual patterns

ADDITIONAL PREPARATION

TEACHER TIPS

MATERIALS

COMMON MISCONCEPTIONS

- Students may have difficulty assessing the reasonableness of their solution.
- Students may cross multiply instead of row multiply.
- Students may multiply the numerator by the denominator when multiplying by a whole number.
- Students may leave the denominator unchanged in fraction multiplication problems.
- Students may believe that multiplication always results in a larger number.

THINGS TO CONSIDER

- Students may know how to perform multiplication abstractly but may not understand the concept of multiplication of fractions. It is suggested that students use models to support their understanding. Supporting their conceptual understanding with models may build a student's ability to assess the reasonableness of an answer.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

1.3.2 Guided Instruction and 1.3.3 Guided Practice position students to notice how fraction multiplication can be represented in multiple ways as well as how these representations illustrate the various strategies that can be used to solve multiplication problems. Students compare how these methods are similar and connect the representations with the algorithm and problem-solving strategies.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

Throughout this lesson, students build understanding through various models and representations, like the array and area model. Students create and find similarities between the models and make connections between the representations. As the lesson progresses, continue to ask students to come up with representations that can help them relate algorithms and equations that students can use to solve a problem.

DIFFERENTIATE INSTRUCTION

Provide students with a variety of tools and strategies for making sense of fractions.

- Consider using graph paper to model the steps to create an array in 1.3.2 Guided Instruction for additional entry points to connect arrays to standard algorithms for multiplication.
- 1.3.1 Warm-Up introduces students to array models through the context of visual patterns. Consider providing math counters or manipulatives for students to recreate the patterns with.
- Consider using questions from 1.3.4 Collaboration for students to write and draw an example of an array and an area model and the corresponding expressions to solve for the product in an interactive math notebook or journal. This reference may be particularly helpful for ELLs to remember vocabulary and the steps for using the algorithm correctly.

1.3.1 WARM-UP: PATTERNS

Component Overview: Students investigate two visual patterns to create an expression to represent each pattern. Be sure students understand that Pattern 1 is the top row and Pattern 2 is the bottom row, with the term numbers written in between both patterns.

TEACHER MOVES

Compare and Connect

As students analyze the two patterns for questions 1 and 2, encourage students to compare similarities and connect the patterns to array models (which will be used in this lesson) by asking:

- What do you notice about how the terms change in Pattern 1? In Pattern 2?
- What is interesting about Term 1 and 2 for both patterns? Why do you think the pattern changes at Term 3?
- How could you use multiplication to represent the different terms? What representation of multiplication does this remind you of?

DIFFERENTIATED INSTRUCTION

Consider having math counters or manipulatives available for students to recreate and extend both patterns. This kinesthetic and visual approach to seeing the pattern as a growing model may help students identify how the pattern is changing from term to term and connect the representations to multiplication/array models.

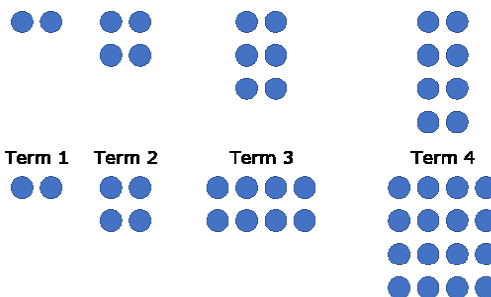
TEACHER MOVES

Students may struggle with writing an expression for Pattern 2. Students may write an expression that relies on the previous term such as 2 times the term before. Students could also write the expression 2^x , but for most students an exponential expression is beyond their mathematical understanding at this point.



1.3.1 Warm-Up: Patterns

Pattern 1



Pattern 2

1. What do you notice about the terms in Pattern 1?
Answers will vary. Each term increases by 2.
2. What do you notice about the terms in Pattern 2?
Answers will vary. Each term doubles.
3. Write an expression to represent each pattern. Use x for the term number.

Pattern 1	Pattern 2
$2x$	2^x



1.3.2 Guided Instruction: Multiplying Fractions

Alejandro took a trip to Key West, Florida. While there, he ate 2 slices of Key Lime Pie. He enjoyed the pie so much that he decided to make some pies for his family. The ingredients for one Key Lime Pie are shown.

Key West's Famous Key Lime Pie		
1 graham cracker crust	$\frac{1}{2}$ cup sour cream	1 tablespoon grated lime zest
3 cups sweetened condensed milk	$\frac{3}{4}$ cup key lime juice	

1. Alejandro wanted to make 5 Key Lime Pies to share with his family.
 - a. Write an expression that can be used to determine how much sour cream Alejandro should buy.
 $5 \times \frac{1}{2}$
 - b. Multiplication can be thought of as

☐ repeated addition.
☐ repeated subtraction.
 - c. Rewrite your expression using the operation selected.
 $\frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2}$
 - d. Draw a model to represent the expression.
Answers will vary.

1.3.2 GUIDED INSTRUCTION: MULTIPLYING FRACTIONS

Component Overview: This Guided Instruction introduces multiplying fractions in the real-world context of making pies by creating visual, area, and array models. Students then use algorithmic approaches based in the visual models to solve multiplication problems.

ENRICHMENT

In question 1, students may not recognize that the recipe components are being multiplied by five. Consider asking students:

- How can we tell what value needs to be multiplied?
- Why do we need to multiply by five to know how much sour cream Alejandro should buy?

DIFFERENTIATED INSTRUCTION

Consider providing fraction manipulatives, like a fraction bar or circle, to represent the fractions in this context. This visual support may elicit student recognition on how to repeatedly draw or represent $(\frac{1}{2} \times 5)$ for question 1. All students will also benefit from understanding the connection of the visual models to the expression rather than memorizing the steps to the algorithm.

1.3.2 GUIDED INSTRUCTION: MULTIPLYING FRACTIONS (CONTINUED)

TEACHER MOVES

Compare and Connect

Students can compare their models for $\frac{1}{4}$ of the pie, identifying models that are similar and comparing if certain models better represent the situation. Students can then compare different models in 2c that result in the same product.

DIFFERENTIATED INSTRUCTION

If students are unclear about how $\frac{1}{4}$ of the pie relates to splitting the pie into eighths, invite them to draw or use fraction circles to create a separate representation using four slices. Then model how the fourths model and the eighths model are the same value because $\frac{1}{4}$ and $\frac{2}{8}$ are equivalent fractions.

ENRICHMENT

If students are unsure how to create an expression to multiply in question 2e, ask:

- How many parts of the whole pie remain? How can we write that amount as a fraction?

QUESTIONING

For early finishers, consider extending thinking by providing additional practice:

Camila gave a second friend half of the remaining pie. Write a multiplication expression whose value represents the amount of pie that Camila has left.

Prove that your multiplication expression results in the amount of pie that Camila has left by using a different pattern of shading in your model to show how much of the pie the second friend received.

2. Alejandro's grandma, Camila, sliced her pie to share with her friends. She cut the pie into 8 slices.

- a. Draw a model of the pie cut into 8 slices below.



- b. Camila gave her friend Liliana $\frac{1}{4}$ of the pie. Use your model to show how many slices Liliana received.

Answer shown on model.

- c. Complete a multiplication expression whose value is the amount of pie that Liliana received.

$$\begin{array}{|c|} \hline 8 \\ \hline \end{array} \times \begin{array}{|c|} \hline 1 \\ \hline 4 \\ \hline \end{array}$$

- d. How many slices of the pie are left?

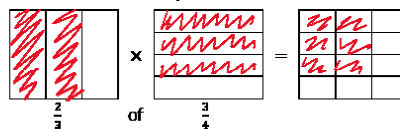
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- e. Complete a multiplication expression whose value represents the amount of pie that remains after Liliana received her slices.

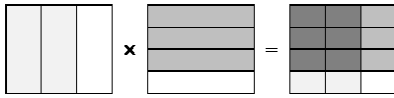
$$\begin{array}{|c|} \hline 8 \\ \hline \end{array} \times \begin{array}{|c|} \hline 3 \\ \hline 4 \\ \hline \end{array}$$

3. Alejandro wanted to make a sixth pie but realized that he only had $\frac{2}{3}$ of the amount of key lime juice that he needed.

- a. Below is an incomplete model that represents the amount of key lime juice that Alejandro has. Shade in the model to represent the context.



- b. The diagram of three figures shows the complete model that represents the amount of key lime juice that Alejandro has. When the first two figures in the diagram are placed on top of each other, the result would be the third figure. In the third figure, the denominator represents the total parts. The numerator is represented by the number of parts where the shading overlaps. Fill in the fractions to complete the equation below the model.



$\frac{2}{3}$ $\times$ $\frac{3}{4}$ = $\frac{6}{12}$

c. How much key lime juice did Alejandro have?
 $\frac{1}{2}$ cup

4. When multiplying fractions, the numerator of the first factor is multiplied by the ☐ numerator ☐ denominator of the second factor. The denominator of the first factor is multiplied by the ☐ numerator ☐ denominator of the second factor. Simply put, when multiplying fractions, multiply ☐ straight across. ☐ diagonally (cross-multiply).

1.3.2 GUIDED INSTRUCTION: MULTIPLYING FRACTIONS (CONTINUED)

DIFFERENTIATED INSTRUCTION

While working on question 3b, consider using graph paper to model the steps to create an array. While modeling, ask students to relate the drawing to these questions:

- What is the total number of parts in each fraction?
- What do the shaded parts of the rows represent in the fraction?
- What does the darker shaded part represent?
- What are the double shaded parts representing?

TEACHER MOVES

Compare and Connect

Use the models in question 3b to specifically target the relationship between multiplication and common denominators, noting the relationship between $\frac{2}{3}$ and $\frac{3}{4}$ as twelfths. Additionally, students can make connections between strategies like the array model (rows and columns) and standard algorithms of multiplication.

DIFFERENTIATED INSTRUCTION

After students attempt to complete the statements in question 4 individually, prompt students to read their responses out loud to a partner. This auditory practice may help support students' conceptual understanding and clarify any mistakes with the help of a partner.

1.3.2 GUIDED INSTRUCTION: MULTIPLYING FRACTIONS (CONTINUED)

TEACHER MOVES

As students continue to solve problems in real-world contexts, ask students to consider the reasonableness of their solutions as a check on understanding.

DIFFERENTIATED INSTRUCTION

Some students may need support on using the area model in question 6a. Consider modeling how to split the two parts of the mixed number into the whole and fraction part and write it vertically on the model. This will reinforce the idea that a mixed number's parts can be combined since it represents addition of a whole number and a fraction part.

QUESTIONING

Ask early finishers to create an area model with the rows of 2 and $\frac{4}{5}$ switched. Do you get the same answer? Why or why not?

5. Alejandro knows he needs $7\frac{1}{2}$ cups of key lime juice to make a batch of 10 pies. He wants to make $2\frac{4}{5}$ batches for an upcoming community bake sale.

Complete the statement about how many cups of key lime juice Alejandro needs to make the pies for the bake sale.

To find the total number of cups of key lime juice

he needs, Alejandro should

- ☐ add
☒ multiply

with $7\frac{1}{2}$ because he needs

- ☒ $2\frac{4}{5}$
☐ 10

batches.

6. When multiplying using mixed numbers, an area model can be used.

- a. Complete the left side of the area model with the missing factors, and then find the product for each box.

	7	$\frac{1}{2}$
2	14	$\frac{2}{2}$
$\frac{4}{5}$	$\frac{28}{5}$	$\frac{4}{10}$

- b. Add the values from each box together to find the total number of cups of key lime juice.
21 cups of key lime juice



1.3.3 Guided Practice: Multiplying Fractions

1. Find the product.

a. $6 \times \frac{3}{5} = \frac{18}{5}$ or $3\frac{3}{5}$

b. $\frac{7}{10} \times \frac{1}{2} = \frac{7}{20}$

2. Evaluate each of the expressions using the area model.

a. $2\frac{3}{5} \times 4\frac{1}{3} = 11\frac{4}{15}$

	4	$\frac{1}{3}$
2	8	$\frac{2}{3}$
$\frac{3}{5}$	$\frac{12}{5}$	$\frac{3}{15}$

b. $4\frac{7}{8} \times 6\frac{1}{4} = 30\frac{15}{32}$

	6	$\frac{1}{4}$
4	24	$\frac{4}{4}$
$\frac{7}{8}$	$\frac{42}{8}$	$\frac{7}{32}$

1.3.3 GUIDED PRACTICE: MULTIPLYING FRACTIONS

Component Overview: Students are provided Guided Practice on multiplying fractions by whole numbers and by fractions. Then students practice multiplying mixed-number fractions using the area model.

DIFFERENTIATED INSTRUCTION

For each of the first two problems, encourage students to use whatever strategies they choose from the previous activities to solve these problems. Consider having them use at least two different strategies to prompt students to check for reasonableness of solutions.

ENRICHMENT

To encourage thinking about the reasonableness of the product when multiplying fractions, ask questions like:

- Does your model represent the expression?
- How does the area model help you solve your expression?
- How can you tell your solution is reasonable?
- Should your solution be less than or greater than one? Why?

1.3.4 COLLABORATION: MULTIPLYING FRACTIONS GREATER THAN ONE USING AN ALGORITHM

Component Overview: Students collaborate to discover different multiplication notations for the multiplying-fractions algorithm and synthesize the process by completing statements.

TEACHER MOVES

Consider giving about 10 minutes for partnerships to complete all questions, then regroup as a class to review solutions to questions 4-7 to check understanding on how to convert mixed numbers to improper fractions and use the algorithm.

ENRICHMENT

Position students to analyze the information in the blue box by asking:

- Why do you think we need to rewrite mixed numbers as fractions greater than one (improper fraction) before using the algorithm shown above? Why is the format of the fraction important?

DIFFERENTIATED INSTRUCTION

For an auditory approach, have partners first try to reason and explain aloud how to rewrite $6\frac{2}{9}$ as a fraction greater than one. After students have discussed the process, then have the partners try to fill in the blanks shown in question 4.

TEACHER MOVES

Throughout the Collaboration, when given the option, encourage students to try solving using both the area model and the algorithm. After sharing, have students share which strategy they prefer to use to solve proper-fraction multiplication problems, like from the Guided Practice, and mixed-number multiplication problems, like from the Collaboration.

QUESTIONING

To foster metacognitive thinking and flexible math approaches, ask students to explain to their partner which method they prefer and why: the algorithm or the area model?



1.3.4 Collaboration: Multiplying Fractions Greater Than One Using an Algorithm

When finding the product, such as $\frac{7}{10} \times \frac{1}{2}$, the algorithm for multiplying, $\frac{a}{b} \times \frac{c}{d}$ is used.



An **algorithm** is a process or a set of rules to be followed when doing calculations.

1. Work with your partner to complete the equation for the algorithm.

$$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$$

2. Another way to represent multiplication is to use a dot.

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{a \cdot c}{b \cdot d}$$

3. Multiplication can also be represented using parentheses.

$$\left(\frac{a}{b}\right)\left(\frac{c}{d}\right) = \frac{(a)(c)}{(b)(d)}$$



The algorithm does not account for mixed numbers. To multiply mixed numbers using the algorithm, the mixed number will have to be rewritten as a fraction greater than 1.

4. Work with your partner to complete the statements that explain how to rewrite $6\frac{2}{9}$ as a fraction greater than 1.

Because $6\frac{2}{9} = 6 + \frac{2}{9}$, $6\frac{2}{9}$ can be rewritten by multiplying $\frac{6}{1}$ by $\frac{9}{9}$ so that 6 is written as an equivalent fraction with a denominator of 9. This results in $\frac{54}{9} + \frac{2}{9}$ whose sum is $\frac{56}{9}$. Therefore, $\frac{56}{9}$ is $6\frac{2}{9}$ rewritten as a fraction greater than 1.

5. Write $8\frac{2}{7}$ as a fraction greater than 1.

$$\frac{61}{7}$$

6. Multiply $3\frac{4}{5} \times 2\frac{1}{3}$ by rewriting $3\frac{4}{5}$ and $2\frac{1}{3}$ as fractions greater than 1.

$$3\frac{4}{5} \times 2\frac{1}{3} = \frac{19}{5} \times \frac{7}{3} = \frac{133}{15} \text{ or } 8\frac{13}{15}$$

7.
 - a. What is the product of $3\frac{4}{5} \cdot 2\frac{1}{3}$?

$$\frac{133}{15} \text{ or } 8\frac{13}{15}$$

- b. Verify your answer by completing the area model.

		3	$\frac{4}{5}$
2		6	$\frac{8}{5}$
$\frac{1}{3}$		1	$\frac{4}{15}$

$$6 + \frac{8}{5} + 1 + \frac{4}{15} = \frac{133}{15} \text{ or } 8\frac{13}{15}$$



1.3.5 Wrap-Up: Ticket Out the Door

1. Complete the following.

a. What is the product of $6\frac{2}{3} \cdot 7\frac{1}{2}$?
50

b. Verify your answer by completing the area model.

		6	$\frac{2}{3}$
7		42	$\frac{14}{3}$
$\frac{1}{2}$		3	$\frac{1}{3}$
$42 + \frac{14}{3} + 3 + \frac{1}{3} = 50$			

2. Dan mixes $\frac{1}{4}$ cup of vinegar with a gallon of water to use as a cleaner. How much vinegar does Dan use if he makes 3 batches of the cleaner?

$$3 \times \frac{1}{4} = \frac{3}{4} \text{ cup}$$

1.3.5 WRAP-UP: TICKET OUT THE DOOR

Component Overview: Students work individually, using all or parts of this exercise as a formative assessment to gather data on student progress from this lesson. Students should not use a calculator to complete the Wrap-Up.

QUESTIONING

Consider additional enrichment opportunities for students to practice or test their abilities.

- Check Your Understanding for lesson-related practice
- Spiral content from Lessons 1-2 with EdgeXL